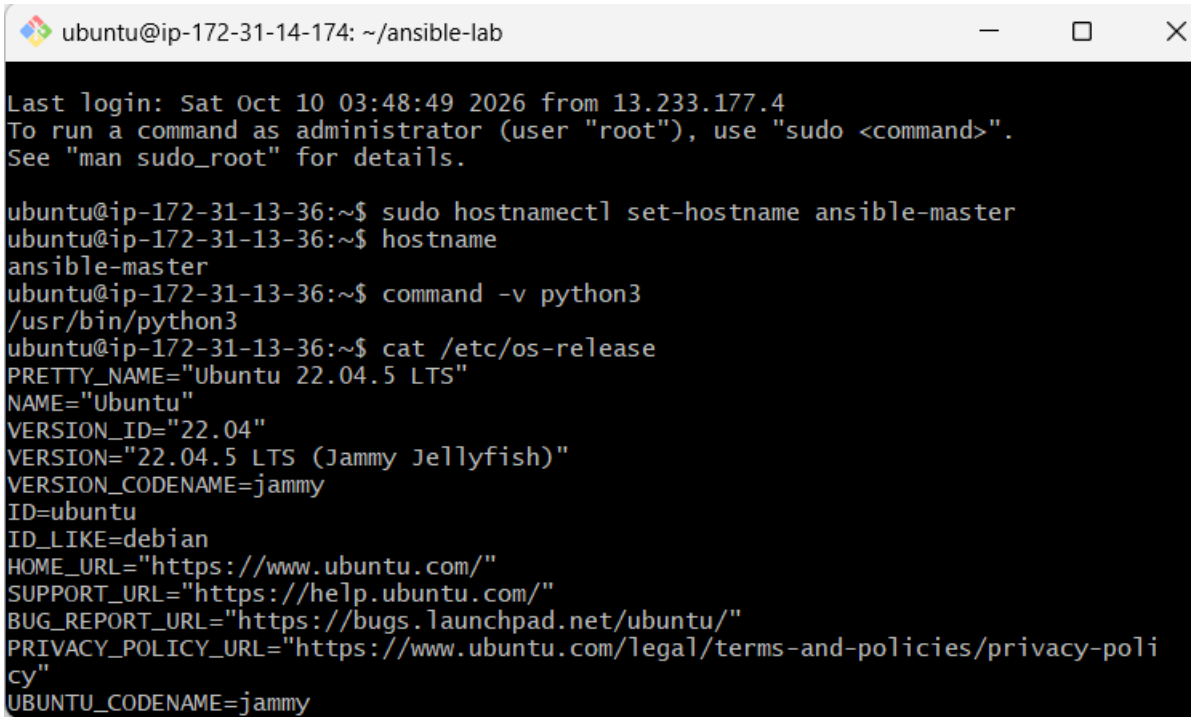
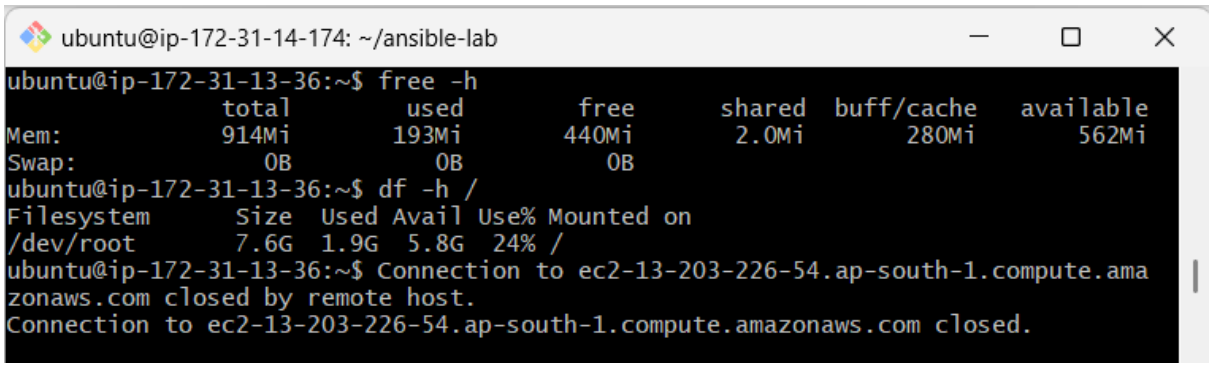
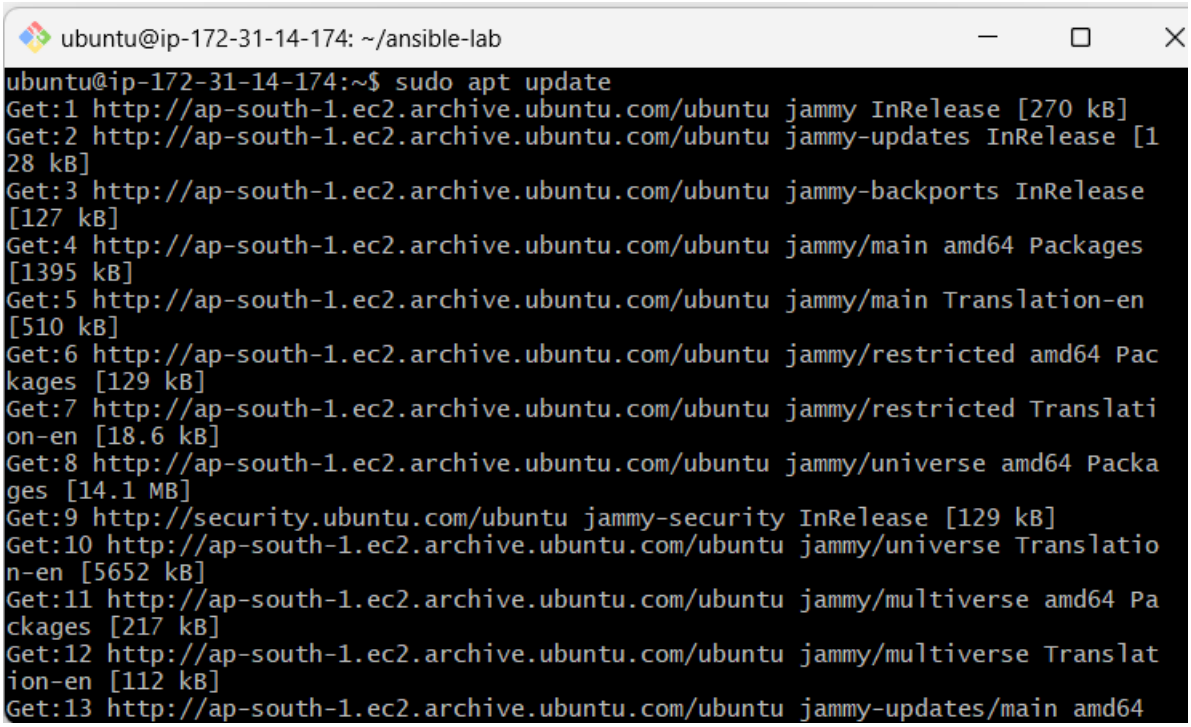
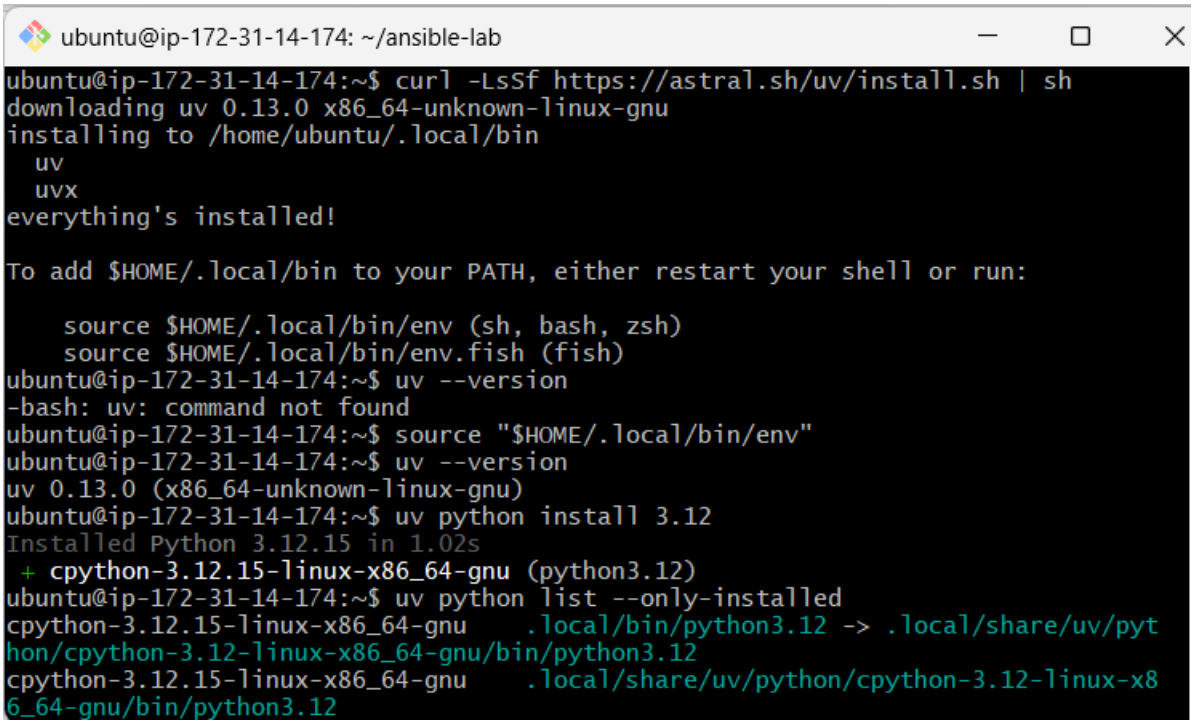
# Ansible Mastery Part-1

## Practical 1: Ansible Installation, Control Node Configuration, Inventory Management, and Ad-Hoc Command Execution

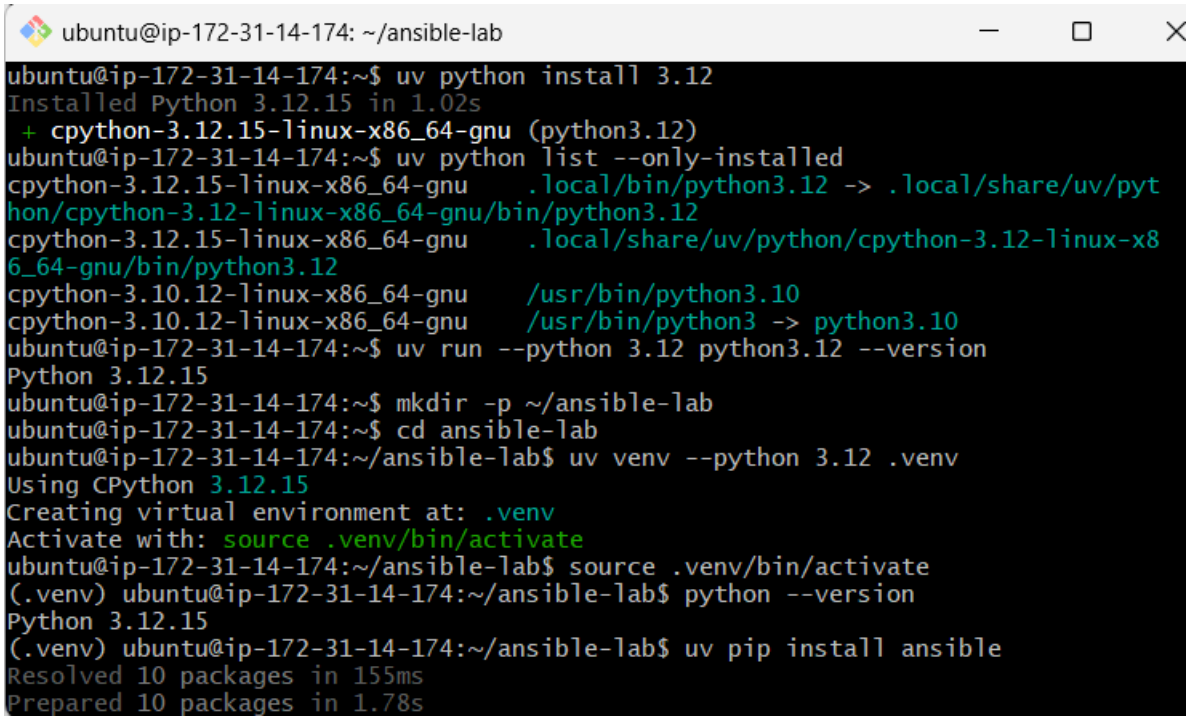


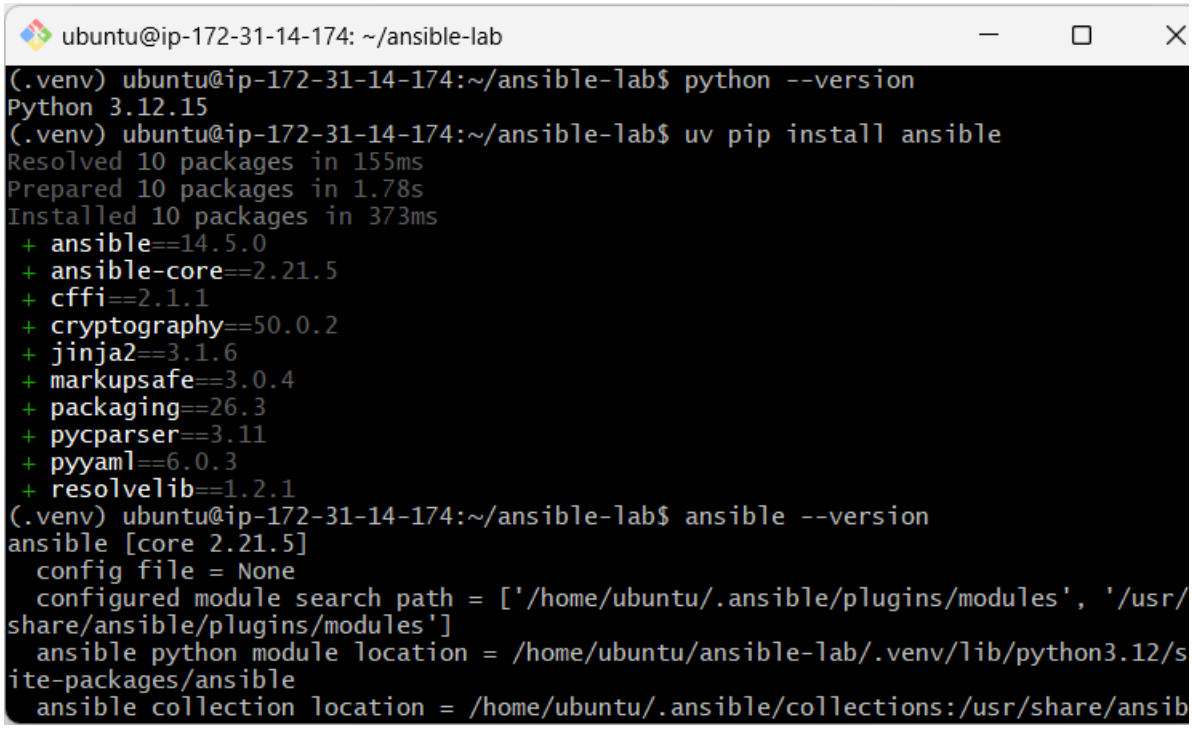


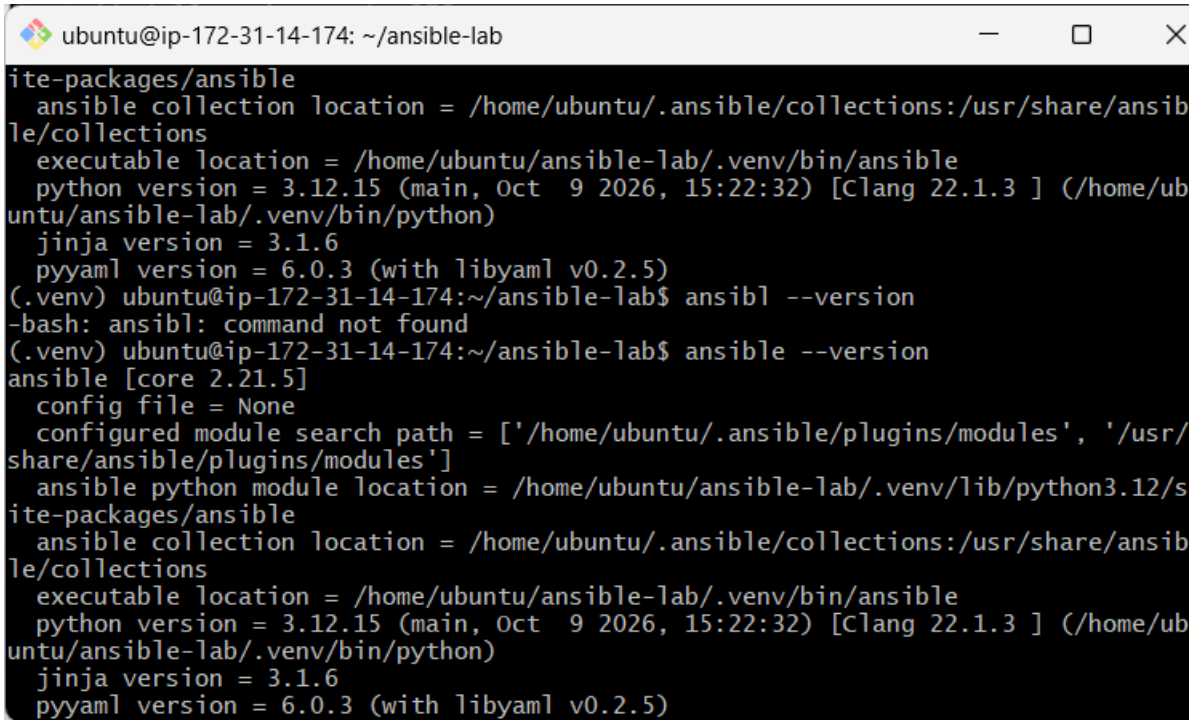



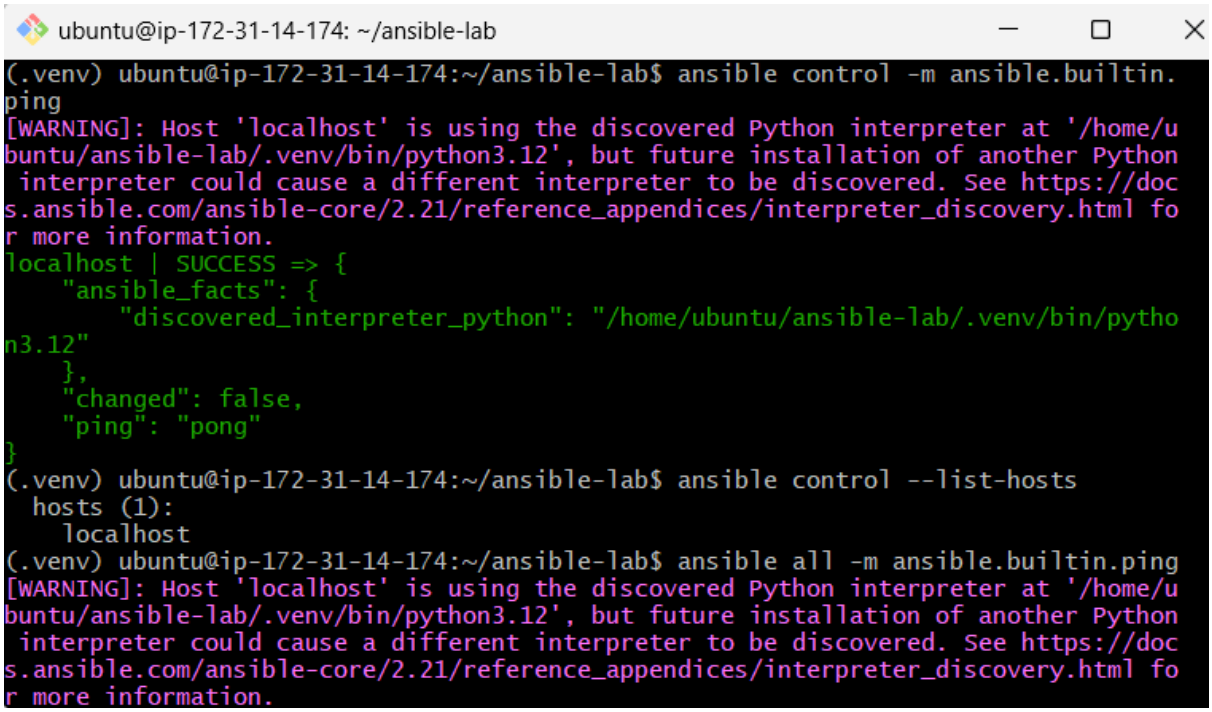

**Note: uv is a fast Python package and project manager that basically simplifies Python installation, dependency management, and virtual env creation.**



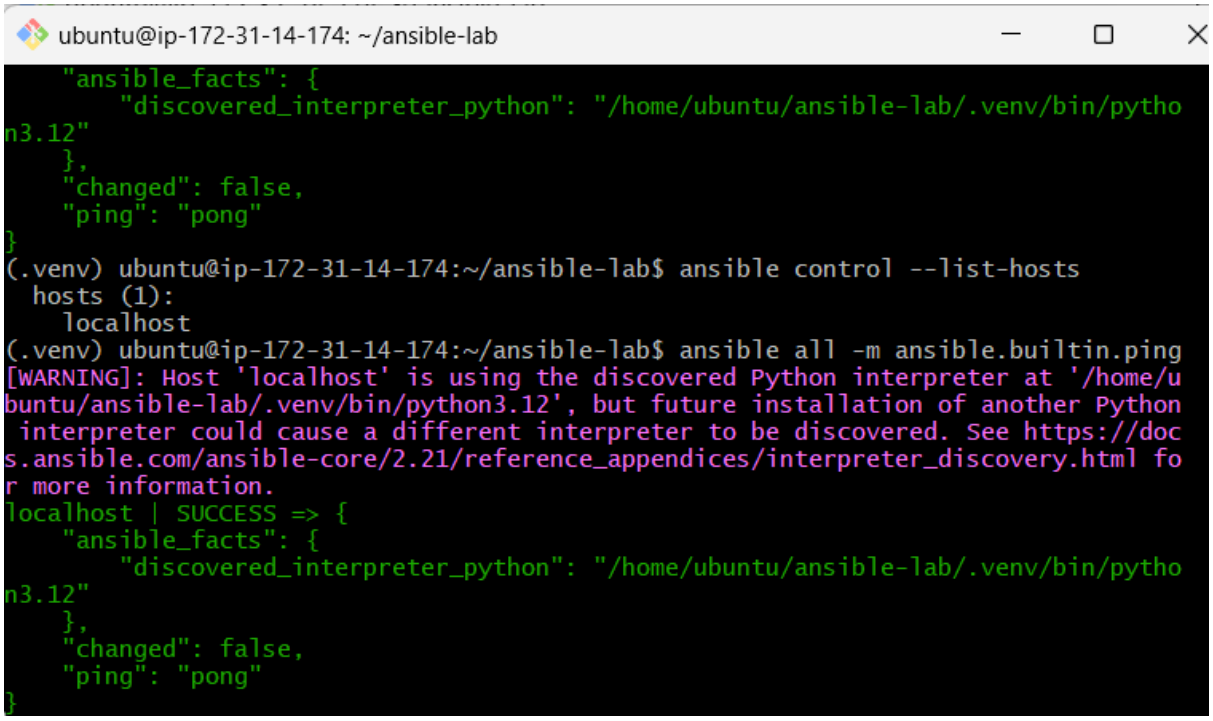

**Note: A Python virtual environment (.venv) isolates project-specific Python packages from the system Python installation, preventing any sort of dependency conflicts. Using uv, we created and activated a Python 3.12 environment, then installed Ansible inside it.**



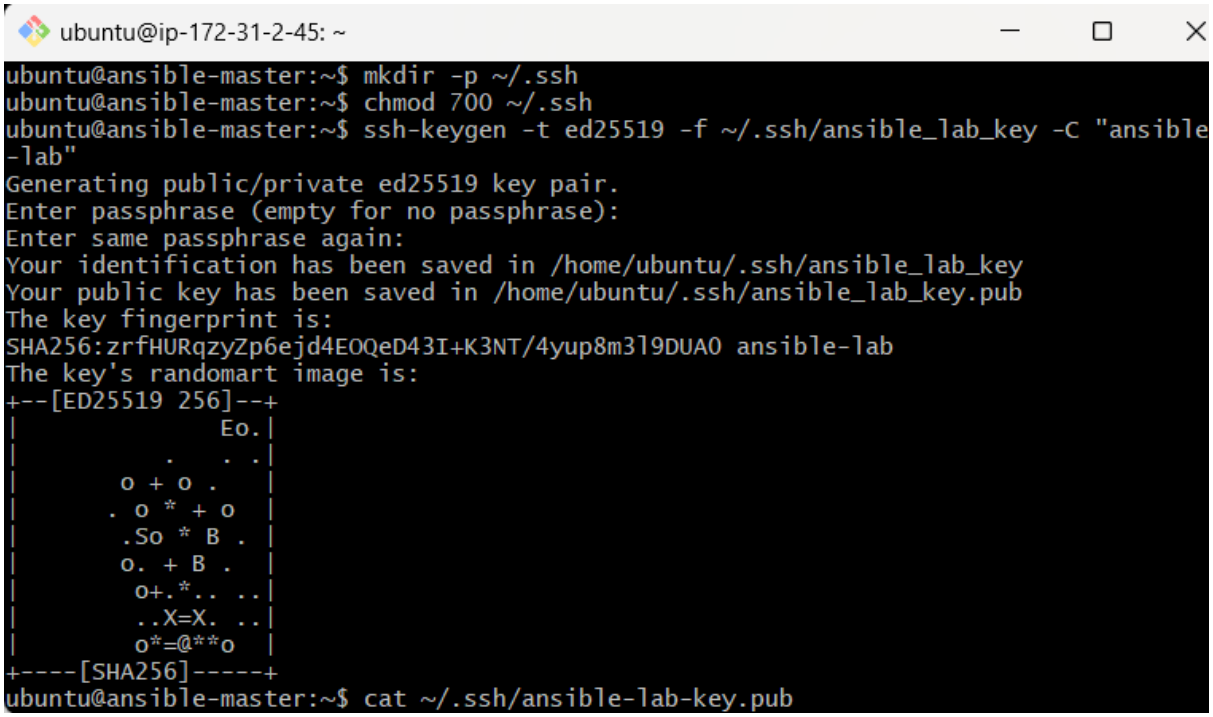

ubuntu@ip-172-31-14-174: ~/ansible-lab
ite-packages/ansible
ansible collection location = /home/ubuntu/.ansible/collections:/usr/share/ansible/collections
executable location = /home/ubuntu/ansible-lab/.venv/bin/ansible
python version = 3.12.15 (main, Oct 9 2026, 15:22:32) [Clang 22.1.3 ] (/home/ubuntu/ansible-lab/.venv/bin/python)
jinja version = 3.1.6
pyyaml version = 6.0.3 (with libyaml v0.2.5)
(.venv) ubuntu@ip-172-31-14-174:~/ansible-lab$ ansibl --version
-bash: ansibl: command not found
(.venv) ubuntu@ip-172-31-14-174:~/ansible-lab$ ansible --version
ansible [core 2.21.5]
config file = None
configured module search path = ['/home/ubuntu/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
ansible python module location = /home/ubuntu/ansible-lab/.venv/lib/python3.12/site-packages/ansible
ansible collection location = /home/ubuntu/.ansible/collections:/usr/share/ansible/collections
executable location = /home/ubuntu/ansible-lab/.venv/bin/ansible
python version = 3.12.15 (main, Oct 9 2026, 15:22:32) [Clang 22.1.3 ] (/home/ubuntu/ansible-lab/.venv/bin/python)
jinja version = 3.1.6
pyyaml version = 6.0.3 (with libyaml v0.2.5)

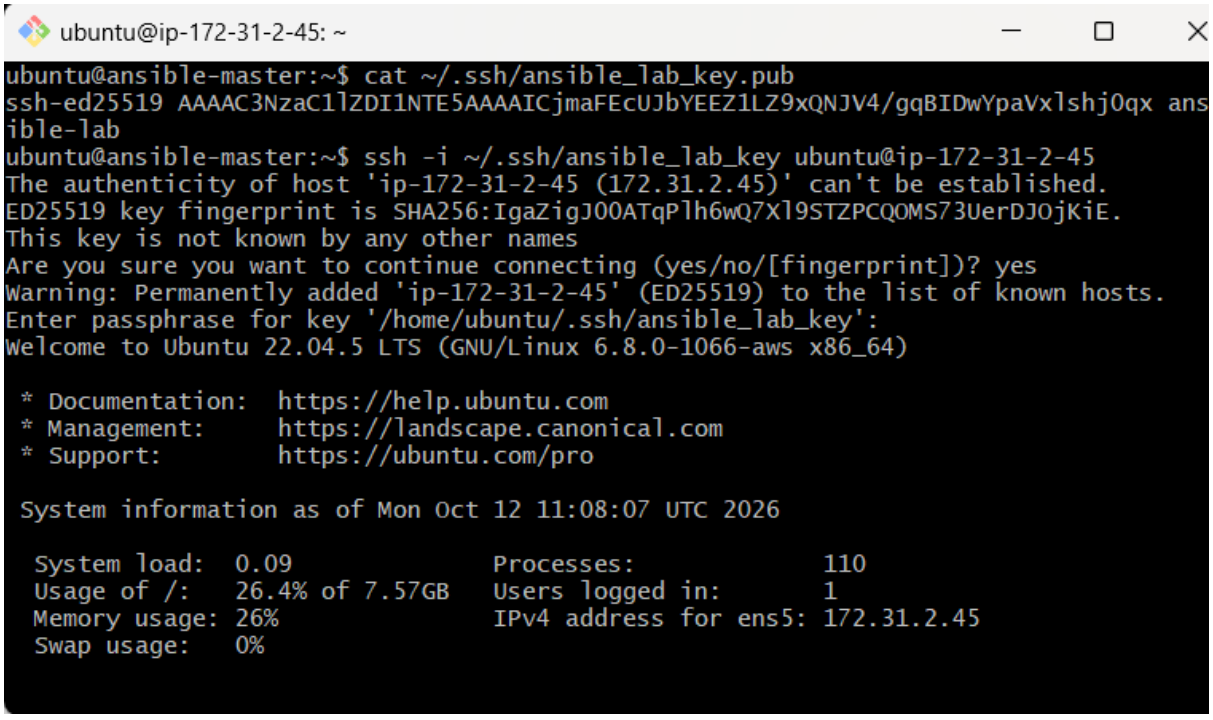

ubuntu@ip-172-31-14-174: ~/ansible-lab
(.venv) ubuntu@ip-172-31-14-174:~/ansible-lab$ ansible control -m ansible.builtin.ping
[WARNING]: Host 'localhost' is using the discovered Python interpreter at '/home/ubuntu/ansible-lab/.venv/bin/python3.12', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.21/reference_appendices/interpreter_discovery.html for more information.
localhost | SUCCESS => {
"ansible_facts": {
"discovered_interpreter_python": "/home/ubuntu/ansible-lab/.venv/bin/python3.12"
},
"changed": false,
"ping": "pong"
}
(.venv) ubuntu@ip-172-31-14-174:~/ansible-lab$ ansible control --list-hosts
hosts (1):
localhost
(.venv) ubuntu@ip-172-31-14-174:~/ansible-lab$ ansible all -m ansible.builtin.ping
[WARNING]: Host 'localhost' is using the discovered Python interpreter at '/home/ubuntu/ansible-lab/.venv/bin/python3.12', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.21/reference_appendices/interpreter_discovery.html for more information.

```
(.venv) ubuntu@ip-172-31-14-174:~/ansible-lab$ cat inventory.ini
[control]
localhost ansible_connection=local
(.venv) ubuntu@ip-172-31-14-174:~/ansible-lab$ cat ansible.cfg
[defaults]
inventory=./inventory.ini
(.venv) ubuntu@ip-172-31-14-174:~/ansible-lab$
```

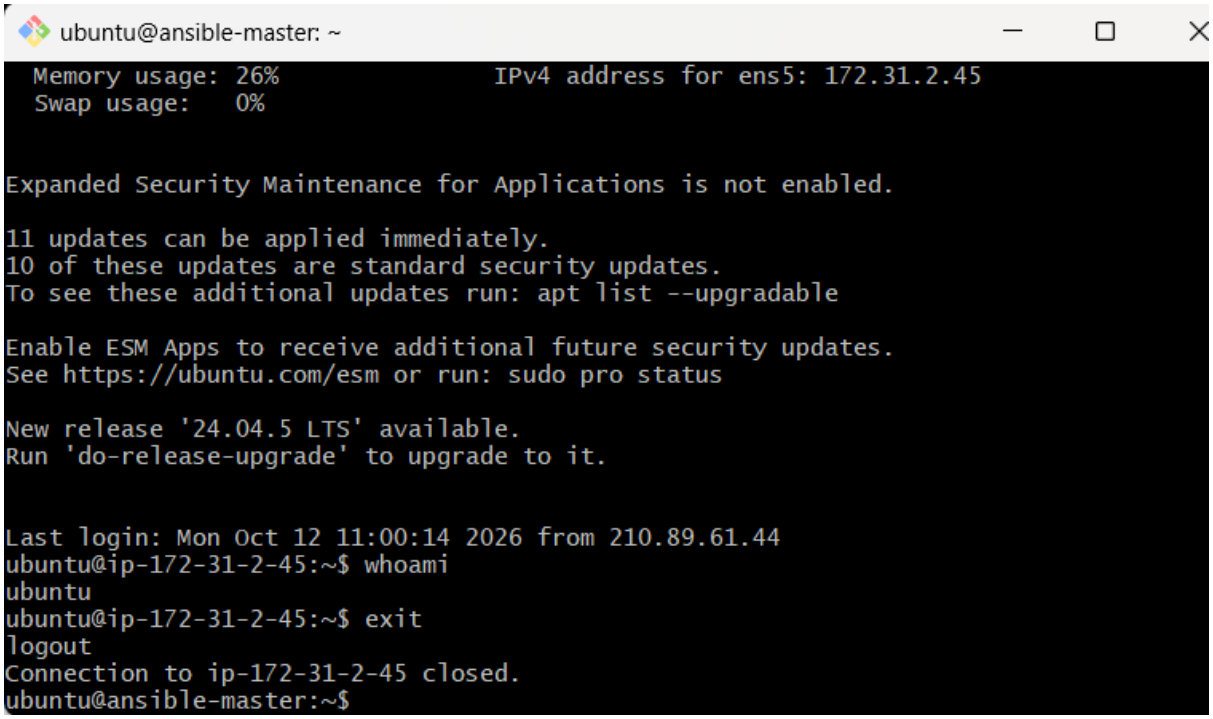


# Practical 2: Remote Node Management Using Ansible - SSH Key-Based Authentication, Inventory Configuration, Ad-Hoc Commands, System Facts, and Privilege Escalation

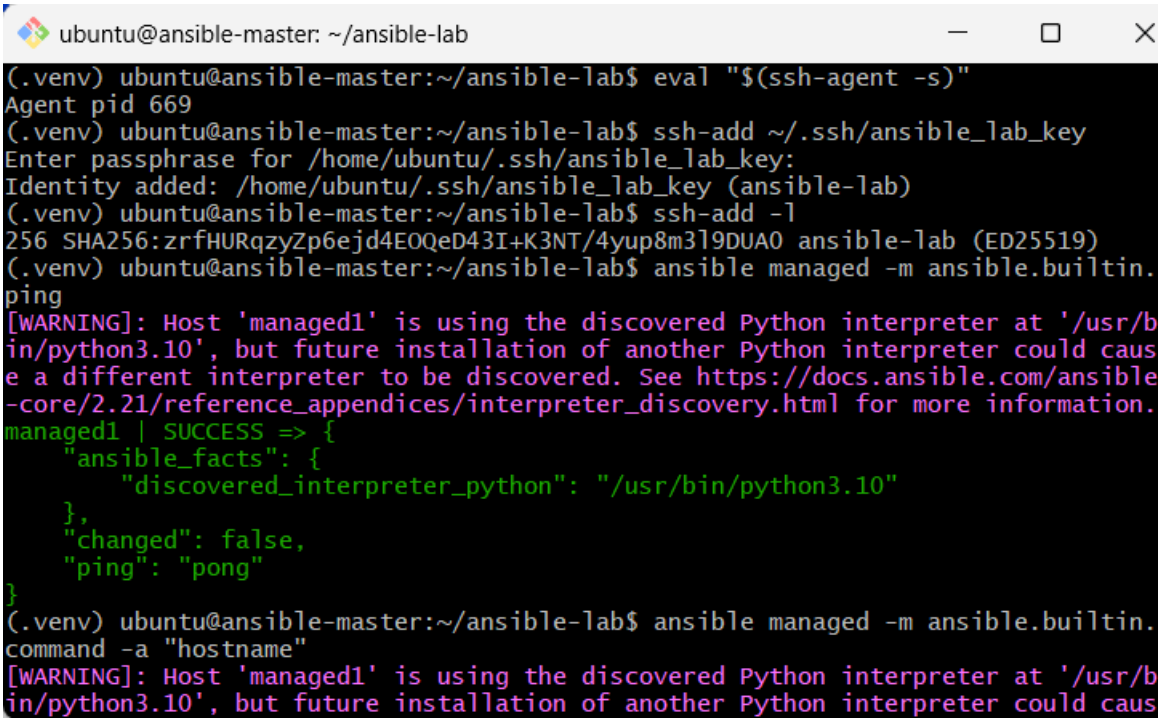


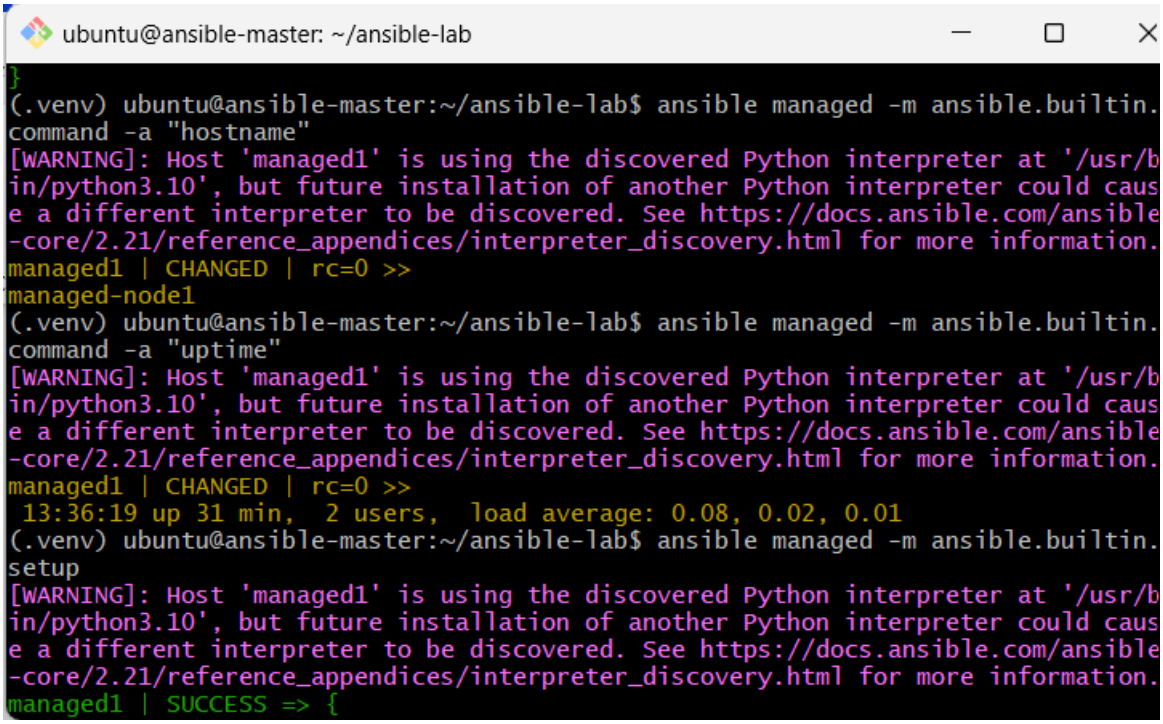

**Note: Do not fail to add the public key of control-node on authorized keys of managed node 1**

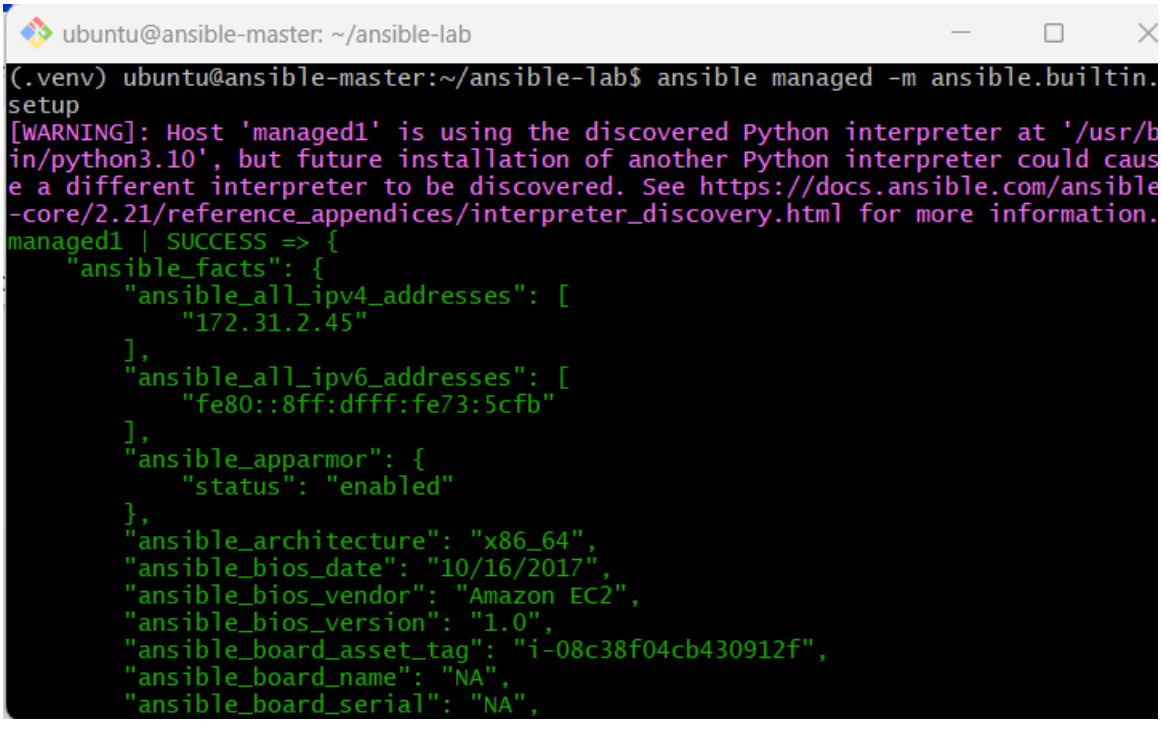



**Note: Successfully connected to managed-node1 via SSH using key-based authentication.**

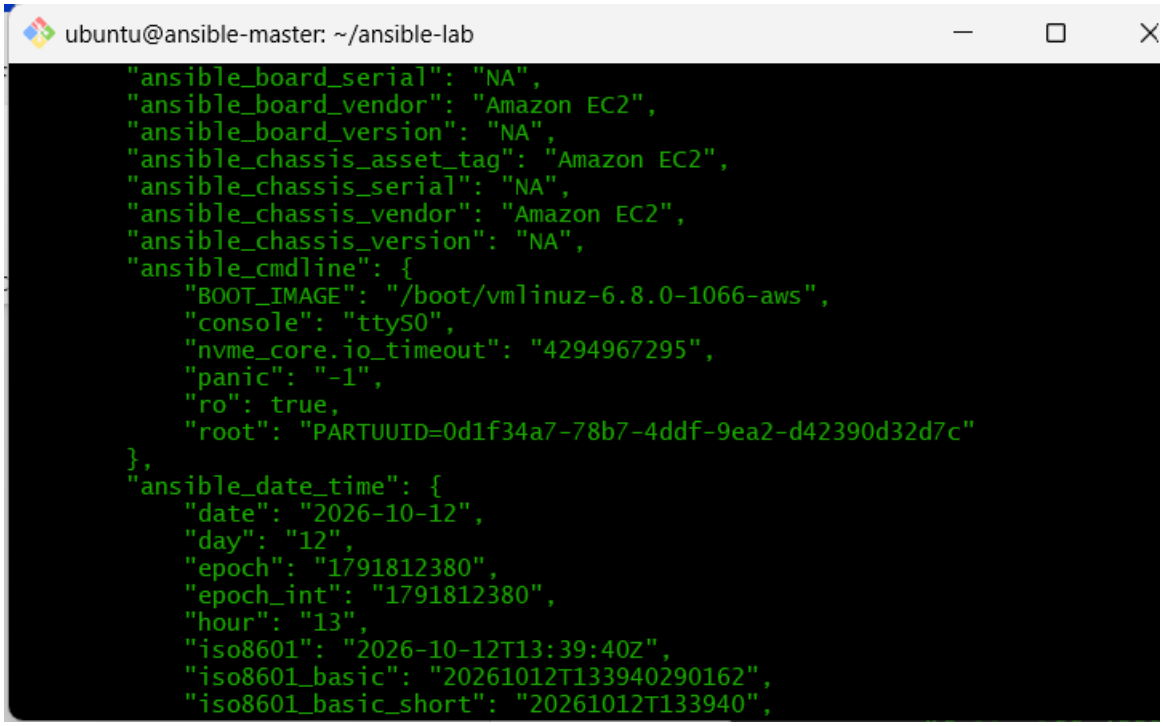


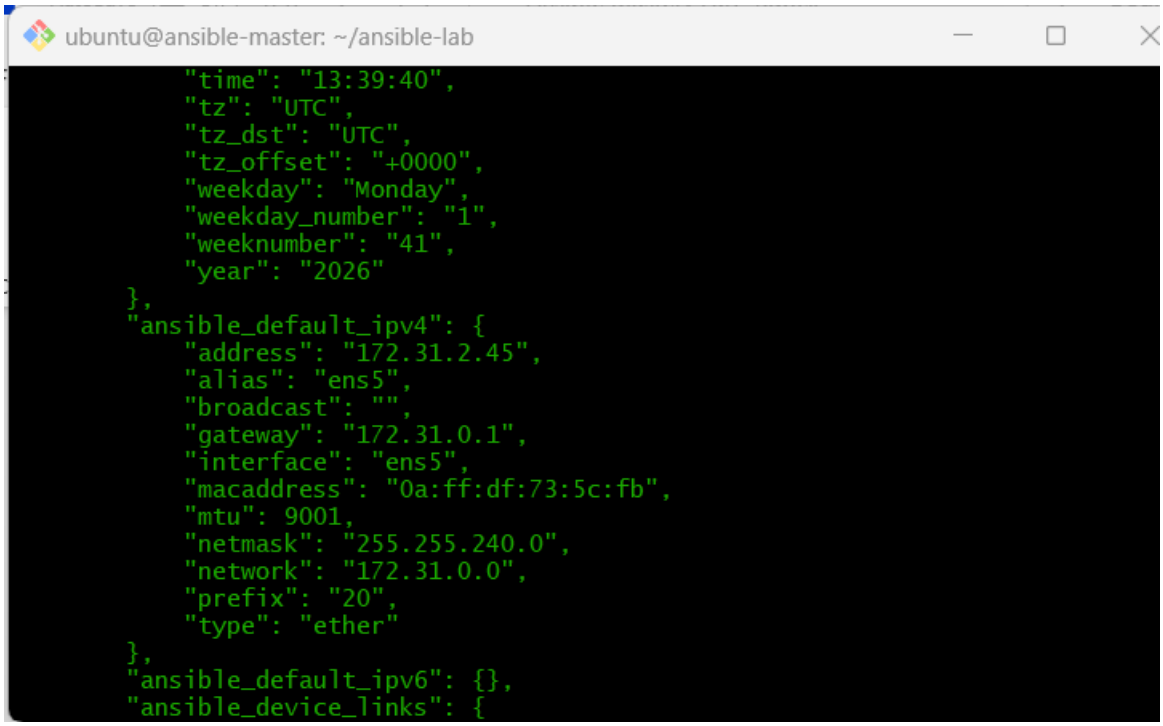

**Note: Configured ssh-agent to securely manage the SSH private key, avoiding repeated passphrase entry during SSH authentication.**

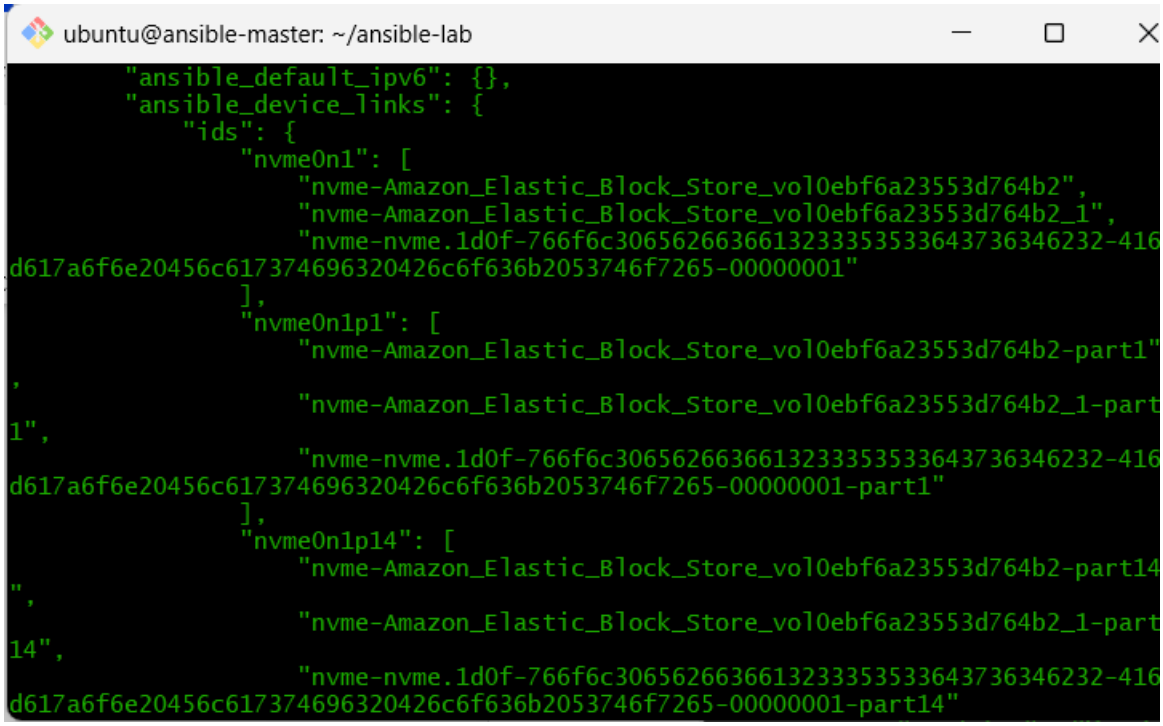


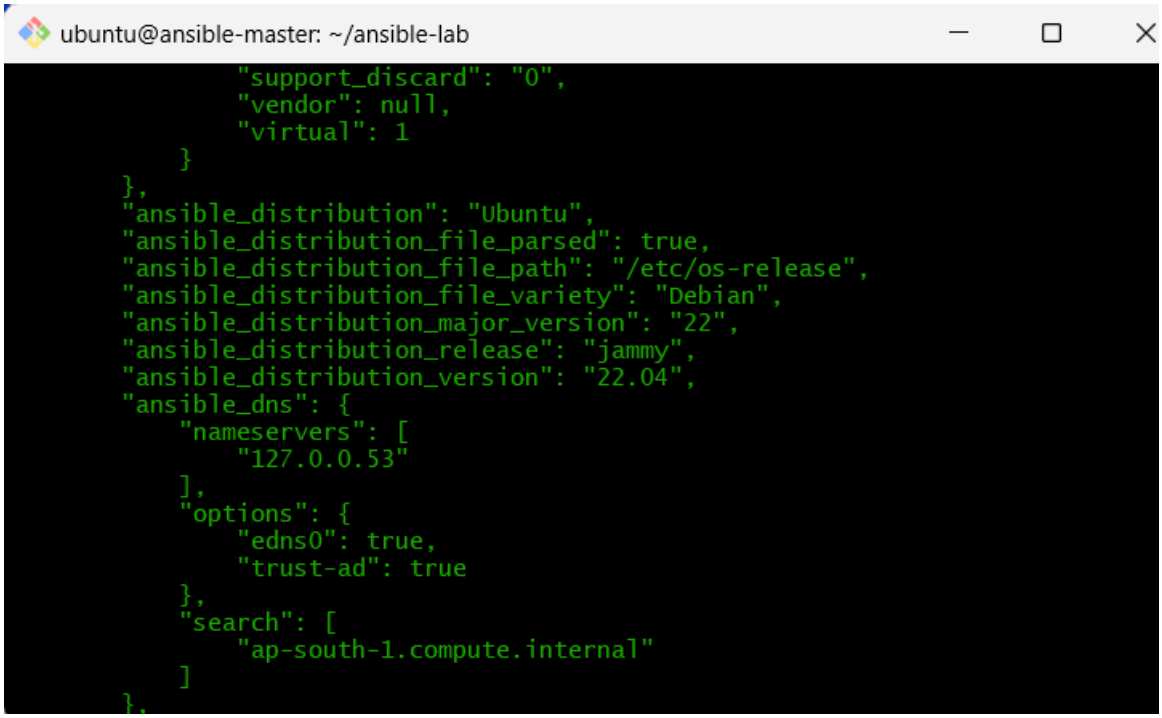



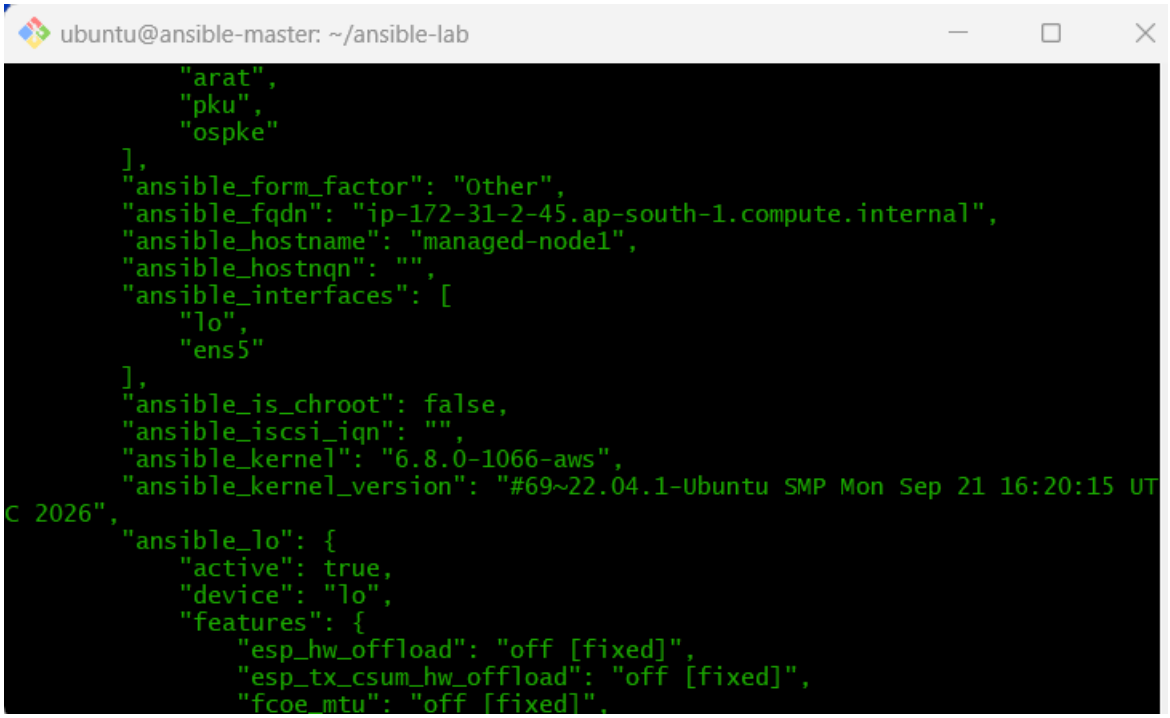

**Successfully gathered system facts from managed-node1 via control node.**

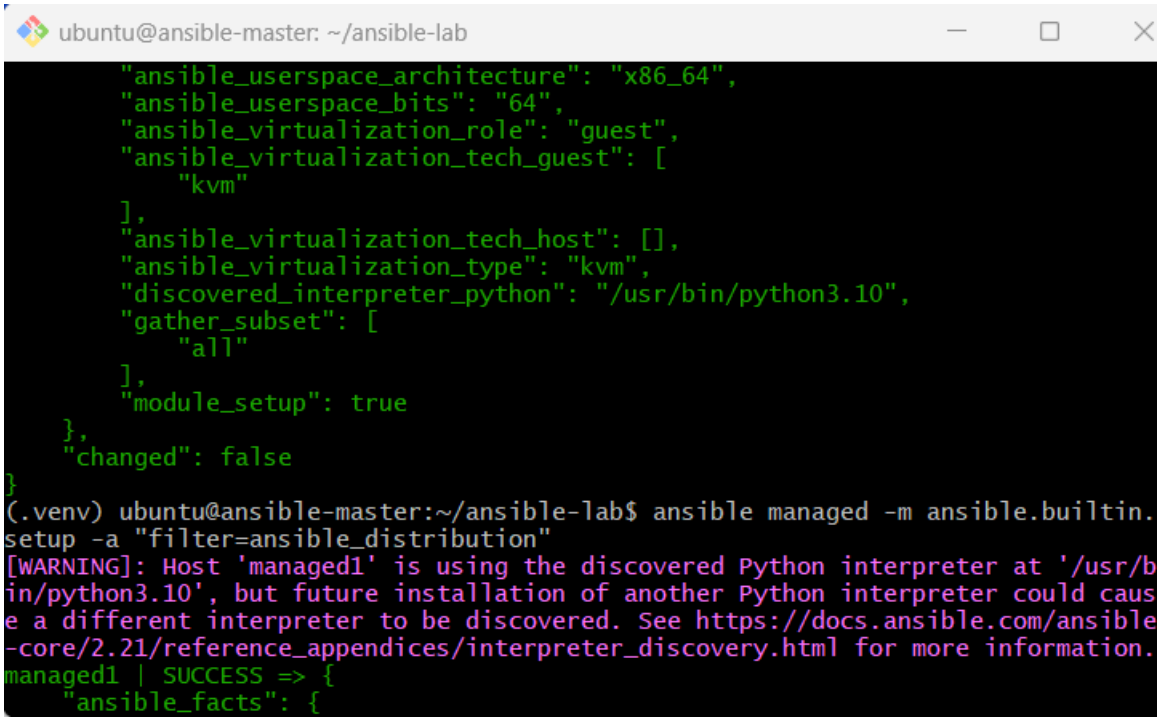

ubuntu@ansible-master: ~/ansible-lab
"ansible_board_serial": "NA",
"ansible_board_vendor": "Amazon EC2",
"ansible_board_version": "NA",
"ansible_chassis_asset_tag": "Amazon EC2",
"ansible_chassis_serial": "NA",
"ansible_chassis_vendor": "Amazon EC2",
"ansible_chassis_version": "NA",
"ansible_cmdline": {
"BOOT_IMAGE": "/boot/vmlinuz-6.8.0-1066-aws",
"console": "ttyS0",
"nvme_core.io_timeout": "4294967295",
"panic": "-1",
"ro": true,
"root": "PARTUUID=0d1f34a7-78b7-4ddf-9ea2-d42390d32d7c"
},
"ansible_date_time": {
"date": "2026-10-12",
"day": "12",
"epoch": "1791812380",
"epoch_int": "1791812380",
"hour": "13",
"iso8601": "2026-10-12T13:39:40Z",
"iso8601_basic": "20261012T133940290162",
"iso8601_basic_short": "20261012T133940",

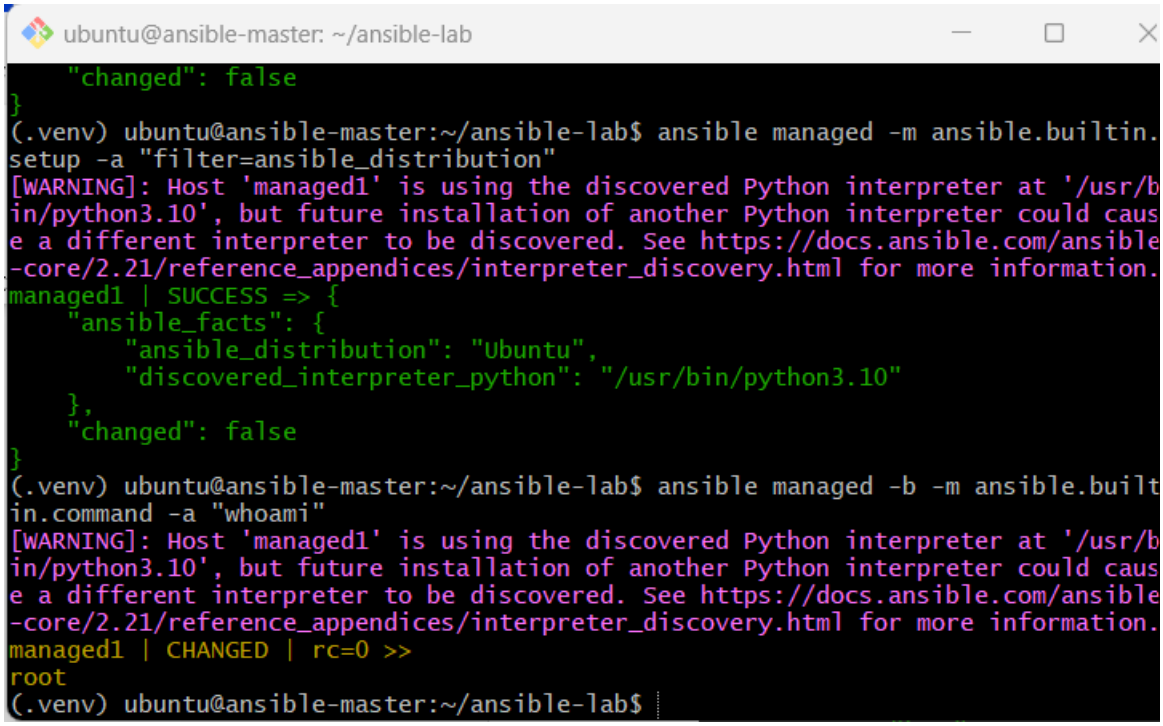

ubuntu@ansible-master: ~/ansible-lab
"time": "13:39:40",
"tz": "UTC",
"tz_dst": "UTC",
"tz_offset": "+0000",
"weekday": "Monday",
"weekday_number": "1",
"weeknumber": "41",
"year": "2026"
},
"ansible_default_ipv4": {
"address": "172.31.2.45",
"alias": "ens5",
"broadcast": "",
"gateway": "172.31.0.1",
"interface": "ens5",
"macaddress": "0a:ff:df:73:5c:fb",
"mtu": 9001,
"netmask": "255.255.240.0",
"network": "172.31.0.0",
"prefix": "20",
"type": "ether"
},
"ansible_default_ipv6": {},
"ansible_device_links": {


ubuntu@ansible-master: ~/ansible-lab
"ansible_default_ipv6": {},
"ansible_device_links": {
"ids": {
"nvme0n1": [
"nvme-Amazon_Elastic_Block_Store_vol0ebf6a23553d764b2",
"nvme-Amazon_Elastic_Block_Store_vol0ebf6a23553d764b2_1",
"nvme-nvme.1d0f-766f6c3065626636613233353533643736346232-416
d617a6f6e20456c617374696320426c6f636b2053746f7265-00000001"
],
"nvme0n1p1": [
"nvme-Amazon_Elastic_Block_Store_vol0ebf6a23553d764b2-part1"
,
"nvme-Amazon_Elastic_Block_Store_vol0ebf6a23553d764b2_1-part
1",
"nvme-nvme.1d0f-766f6c3065626636613233353533643736346232-416
d617a6f6e20456c617374696320426c6f636b2053746f7265-00000001-part1"
],
"nvme0n1p14": [
"nvme-Amazon_Elastic_Block_Store_vol0ebf6a23553d764b2-part14
",
"nvme-Amazon_Elastic_Block_Store_vol0ebf6a23553d764b2_1-part
14",
"nvme-nvme.1d0f-766f6c3065626636613233353533643736346232-416
d617a6f6e20456c617374696320426c6f636b2053746f7265-00000001-part14"


ubuntu@ansible-master: ~/ansible-lab
"support_discard": "0",
"vendor": null,
"virtual": 1
}
},
"ansible_distribution": "Ubuntu",
"ansible_distribution_file_parsed": true,
"ansible_distribution_file_path": "/etc/os-release",
"ansible_distribution_file_variety": "Debian",
"ansible_distribution_major_version": "22",
"ansible_distribution_release": "jammy",
"ansible_distribution_version": "22.04",
"ansible_dns": {
"nameservers": [
"127.0.0.53"
],
"options": {
"edns0": true,
"trust-ad": true
},
"search": [
"ap-south-1.compute.internal"
]
},


ubuntu@ansible-master: ~/ansible-lab
"arat",
"pku",
"ospke"
],
"ansible_form_factor": "Other",
"ansible_fqdn": "ip-172-31-2-45.ap-south-1.compute.internal",
"ansible_hostname": "managed-node1",
"ansible_hostnqn": "",
"ansible_interfaces": [
"lo",
"ens5"
],
"ansible_is_chroot": false,
"ansible_iscsi_iqn": "",
"ansible_kernel": "6.8.0-1066-aws",
"ansible_kernel_version": "#69~22.04.1-Ubuntu SMP Mon Sep 21 16:20:15 UT
C 2026",
"ansible_lo": {
"active": true,
"device": "lo",
"features": {
"esp_hw_offload": "off [fixed]",
"esp_tx_csum_hw_offload": "off [fixed]",
"fcoe_mtu": "off [fixed]",


ubuntu@ansible-master: ~/ansible-lab
"ansible_userspace_architecture": "x86_64",
"ansible_userspace_bits": "64",
"ansible_virtualization_role": "guest",
"ansible_virtualization_tech_guest": [
"kvm"
],
"ansible_virtualization_tech_host": [],
"ansible_virtualization_type": "kvm",
"discovered_interpreter_python": "/usr/bin/python3.10",
"gather_subset": [
"all"
],
"module_setup": true
},
"changed": false
}
(.venv) ubuntu@ansible-master:~/ansible-lab$ ansible managed -m ansible.builtin.
setup -a "filter=ansible_distribution"
[WARNING]: Host 'managed1' is using the discovered Python interpreter at '/usr/b
in/python3.10', but future installation of another Python interpreter could caus
e a different interpreter to be discovered. See https://docs.ansible.com/ansible
-core/2.21/reference_appendices/interpreter_discovery.html for more information.
managed1 | SUCCESS => {
"ansible_facts": {




**There we go, successfully demonstrated privilege escalation using -b flag to execute commands as root on managed node1.**